

Clase 13

Diseño web en la práctica

IIC2513 - Tecnologías y Aplicaciones Web

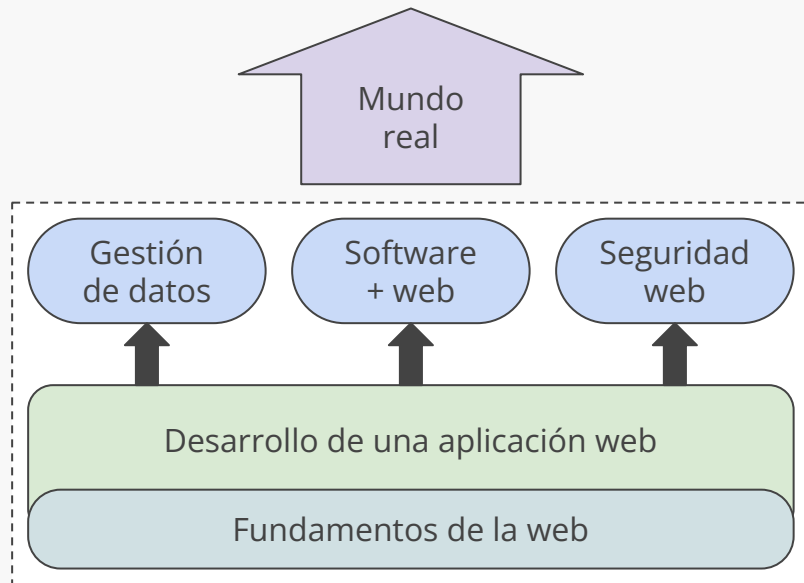
Antonio Ossa Guerra
aaossa@ing.puc.cl

1. Diapositivas charla
2. Evaluación Temprana de Cursos
3. Semana de receso

Anuncios

Contenidos del curso

- (1) Fundamentos de la web
- (2) Desarrollo de una app web
- (3) Gestión de datos
- (4) Software + web
- (5) Seguridad web
- (6) El mundo real

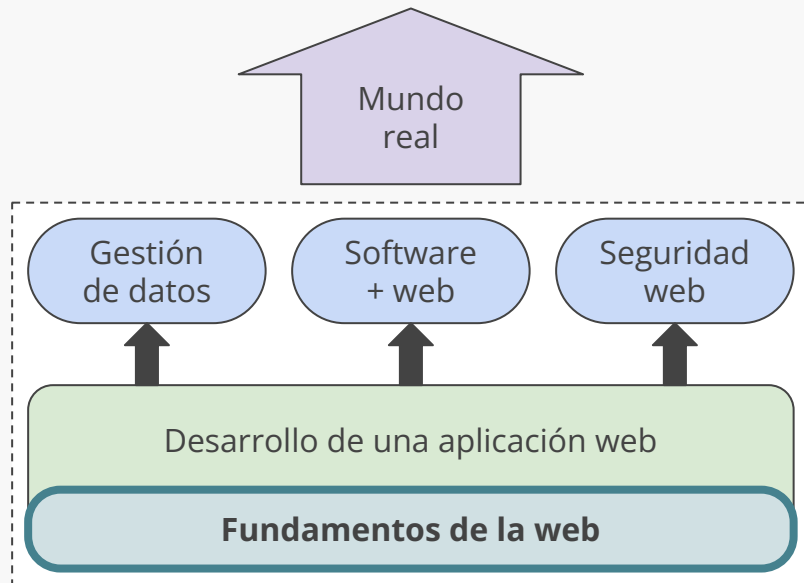


Contenidos del curso

Fundamentos de la web

Veremos definiciones relacionadas con internet, protocolos y qué son **HTML**, **CSS** y **JavaScript**...

... para comprender la web como una **plataforma de desarrollo**, su funcionamiento básico, y protocolos y tecnologías detrás de esta

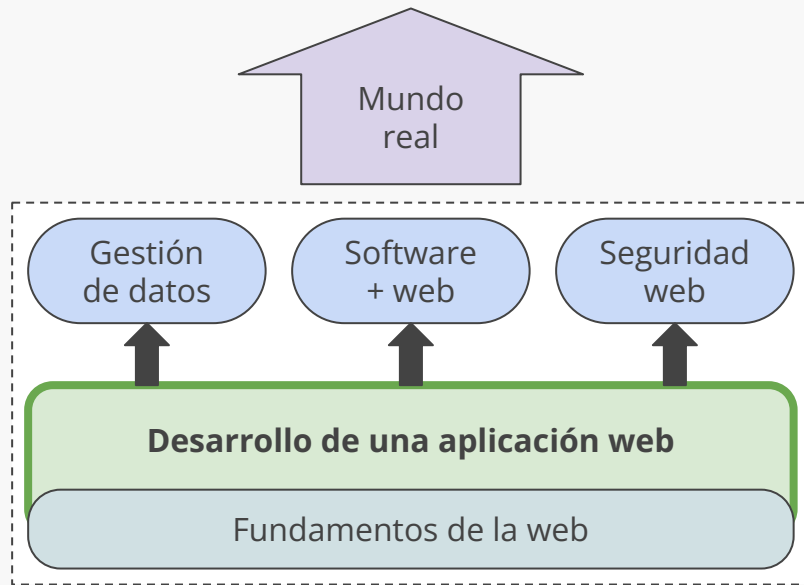


Contenidos del curso

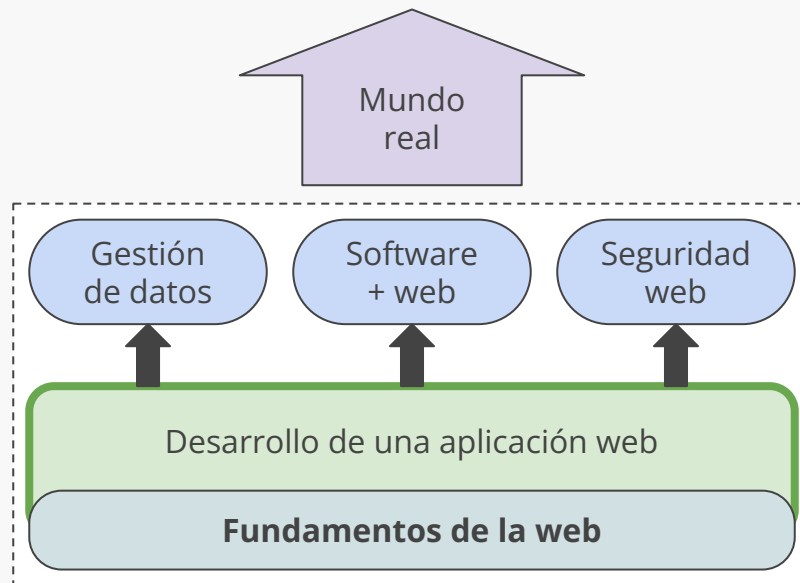
Desarrollo de una aplicación web

Veremos herramientas como React, Node.js, Koa y Express, y técnicas como **CSS avanzado** y el uso de **Design Systems**

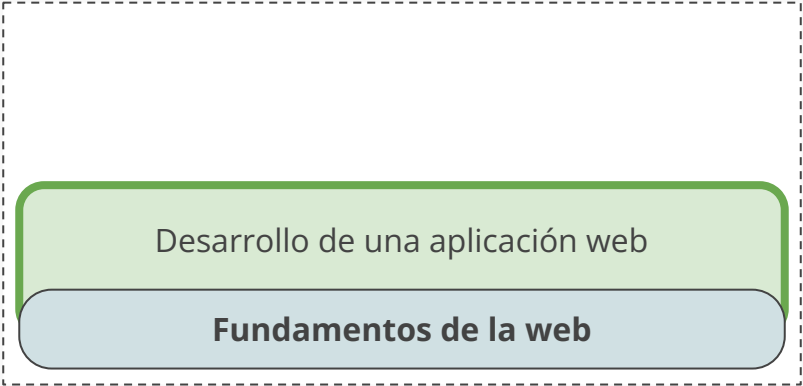
... para desarrollar **aplicaciones modernas** tanto en el lado del **cliente** (front-end) como en el lado del servidor (back-end)



Contenidos del curso



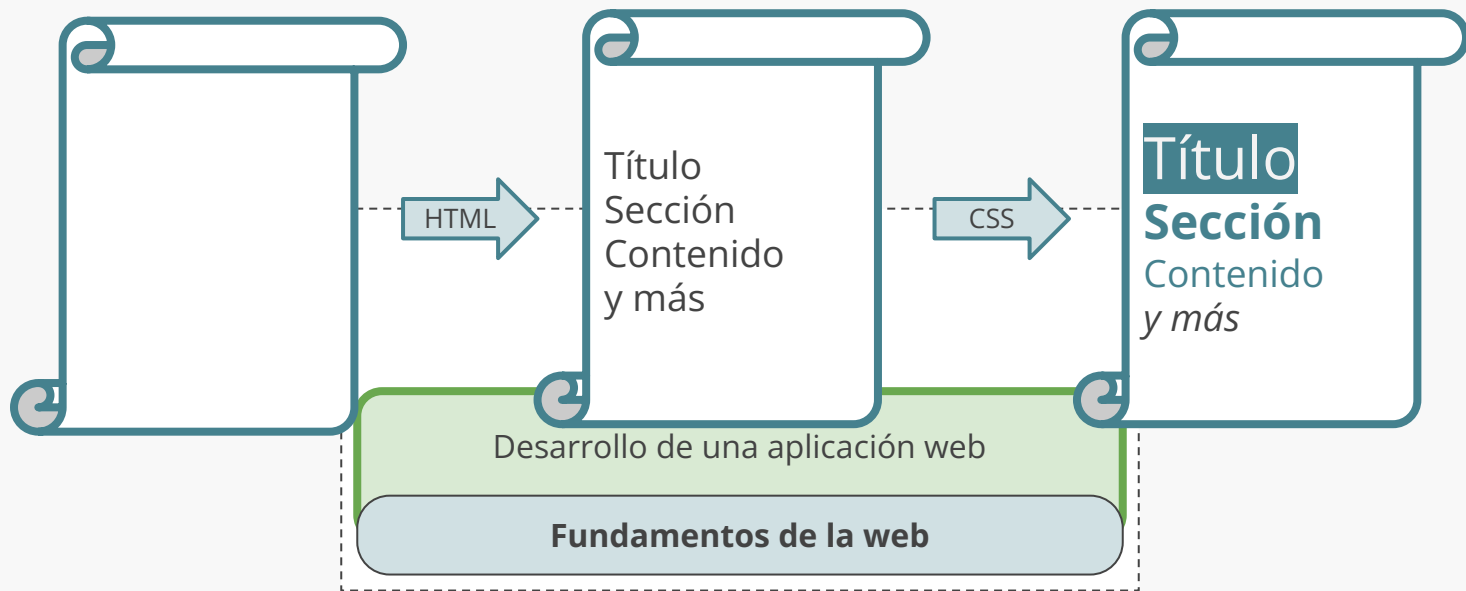
Contenidos del curso



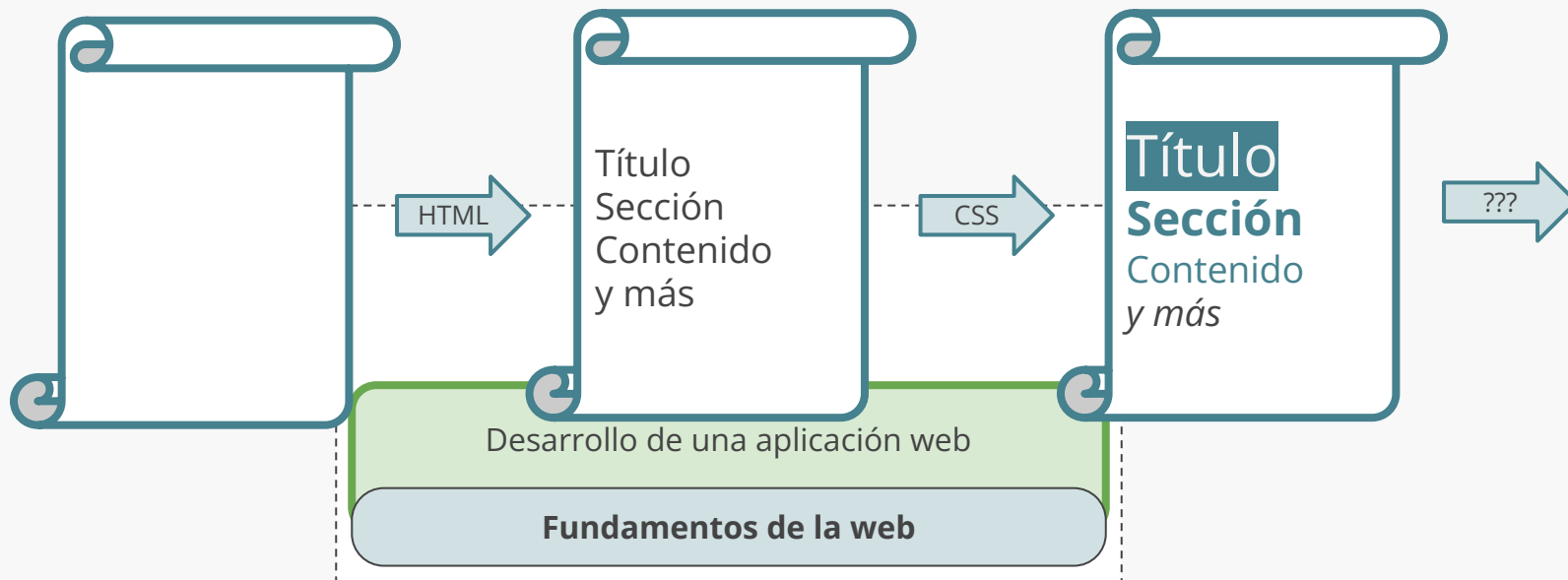
Desarrollo de una aplicación web

Fundamentos de la web

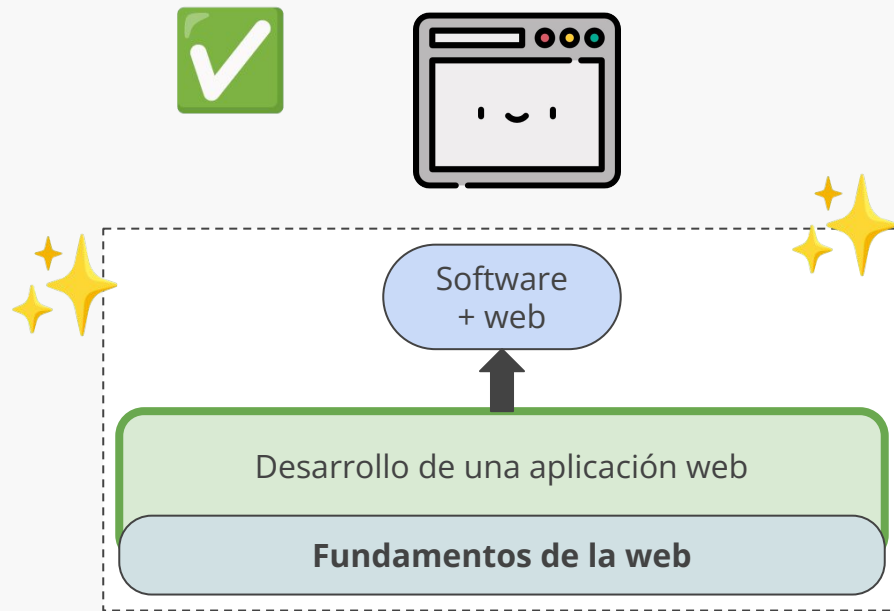
Contenidos del curso



Contenidos del curso



Contenidos del curso



100

Contenidos del curso



Software
+ web



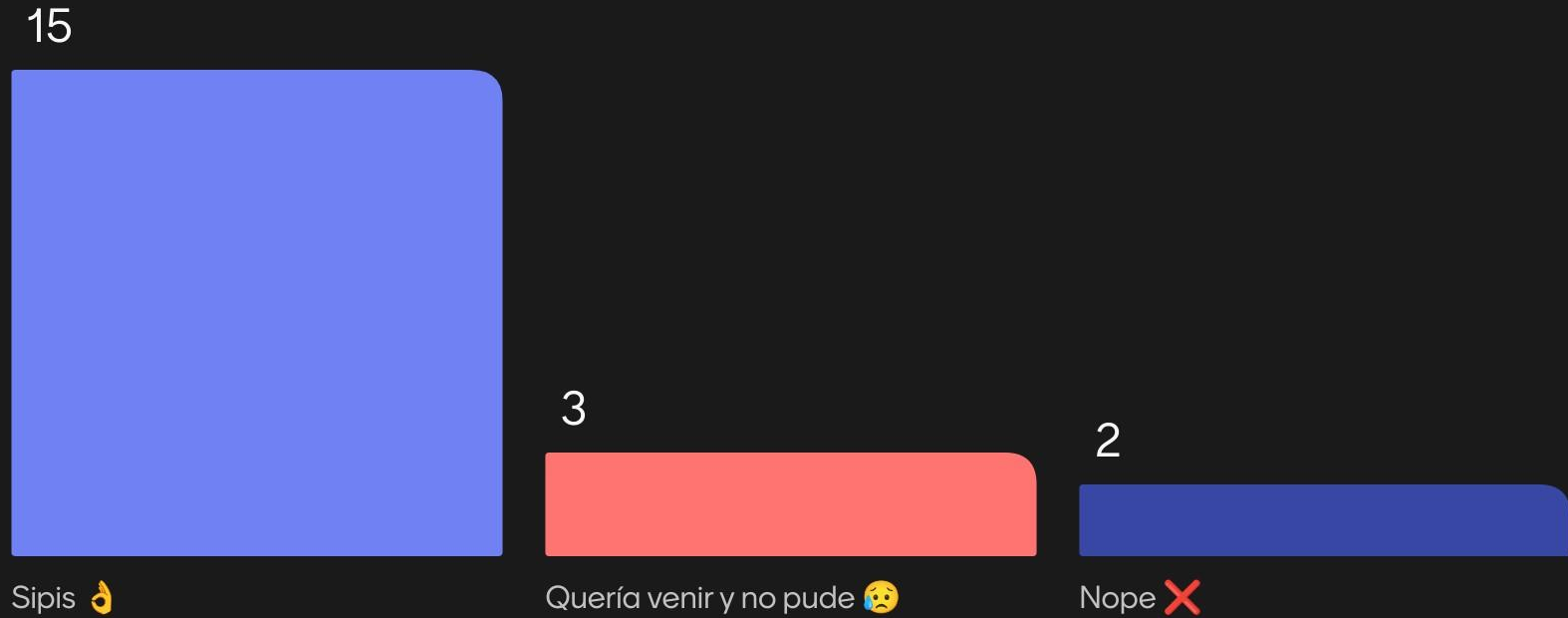
Pregunta# *¿Cómo podemos implementar el diseño?*

Fundamentos de la web

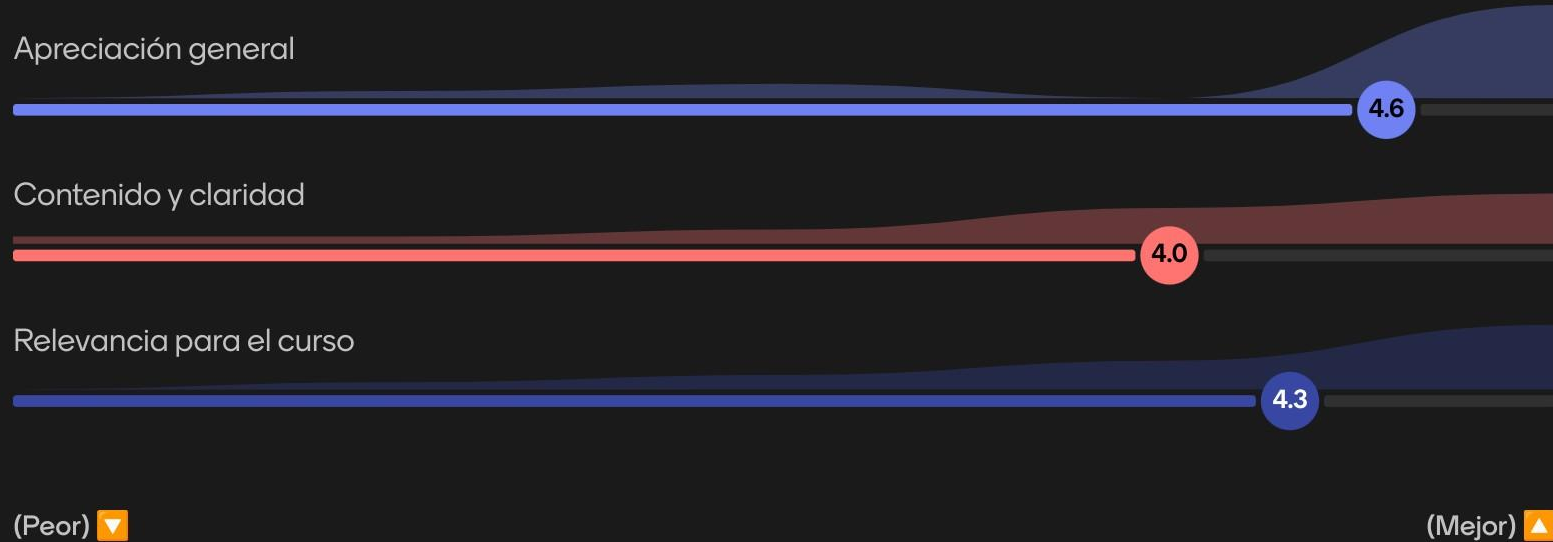
Menti*#Hablemos de diseño web*

Sobre la charla

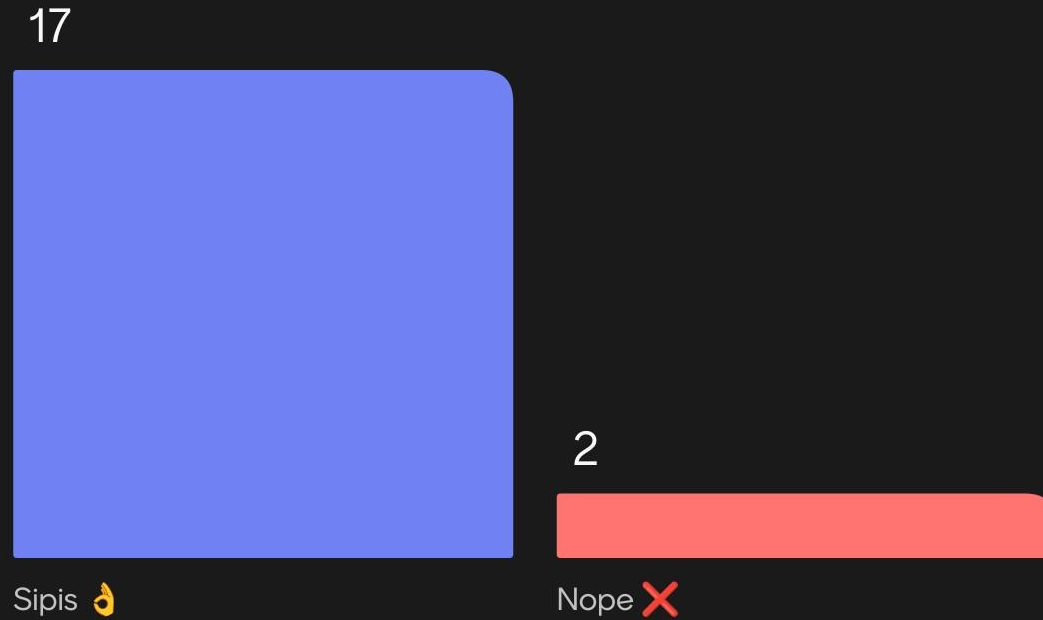
¿Viniste a la charla del martes?



Si viniste a la charla, ¿cómo la calificarías la en...



¿Vendrías a una charla de un próximo invitado en el futuro?



Sobre las próximas semanas

¿Qué harán en su semana de receso? 🤔

36 Respuestas



¿Qué te gustaría ver en la segunda mitad de semestre? 🙄

Un 7

Puros 7

Docker!!!!!!!!!!

decimas

Como hacer una app en
google play

elon musk

Next.js

interrogaciones

¿Qué te gustaría ver en la segunda mitad de semestre? 🙄

Css dinamico

me gustaría poder saber cómo
deployar bien, para poder
empezar a crear mis páginas
web y llevarlas a la realidad ,
eso no se habla mucho

Mobile

La planta

fotos de la planta

La planta

Seguridad

deployar

¿Qué te gustaría ver en la segunda mitad de semestre? 🙄

Principios para mejorar
nuestras aplicaciones web,
sean principios de diseño,
tecnologías útiles y sus
aplicaciones, etc

Next.js

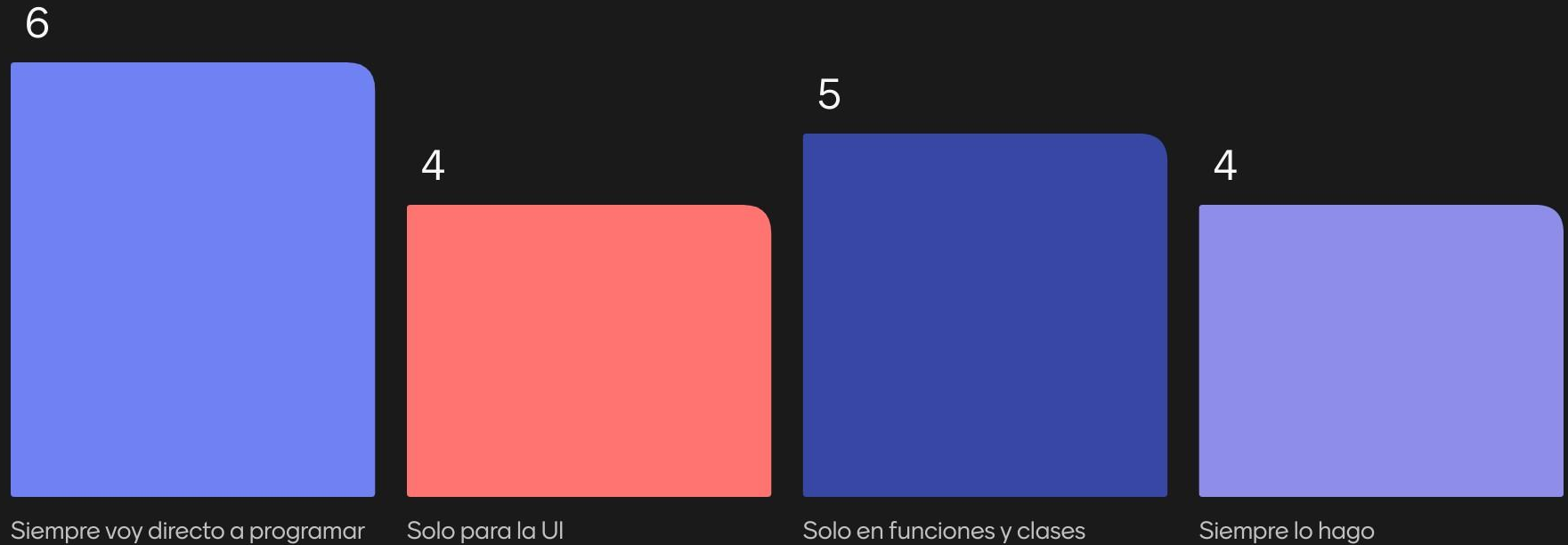
Hablemos de front-end

¿En qué piensas al hablar de "front-end"?

32 Respuestas



¿Qué tan seguido "diseñas" antes de programar?



¡Gracias!

Vamos con esta clase

¿Dónde ya ejecutamos JavaScript?



Google Chrome



Microsoft Edge



Apple Safari



Mozilla Firefox



Opera



Brave

¿Cómo los navegadores ejecutan JavaScript?

V8 JavaScript Engine

[v8/v8 - Git at Google](#)



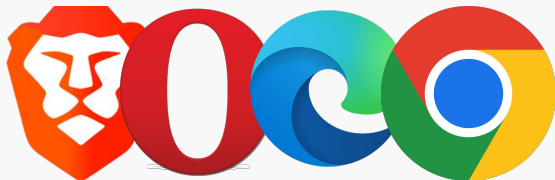
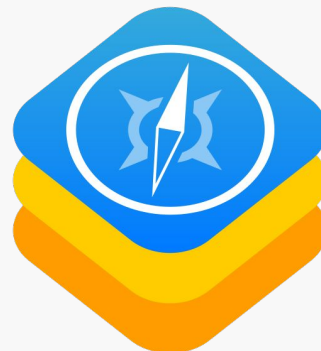
SpiderMonkey

[spidermonkey.dev](#)



JavaScriptCore

[WebKit/WebKit @ GitHub](#)



¿Cómo los navegadores grafican?

Blink

[Blink \(Rendering Engine\)](#)



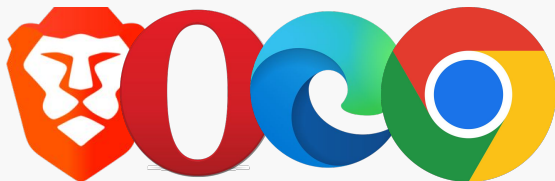
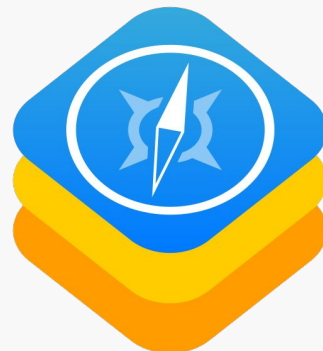
Gecko

[Gecko docs](#)



WebKit

[WebKit/WebKit @ GitHub](#)



Diferencias en *rendering* entre navegadores



Ahmad Shadeed



Ahmad Shadeed

Diferencias en *rendering* entre SOs

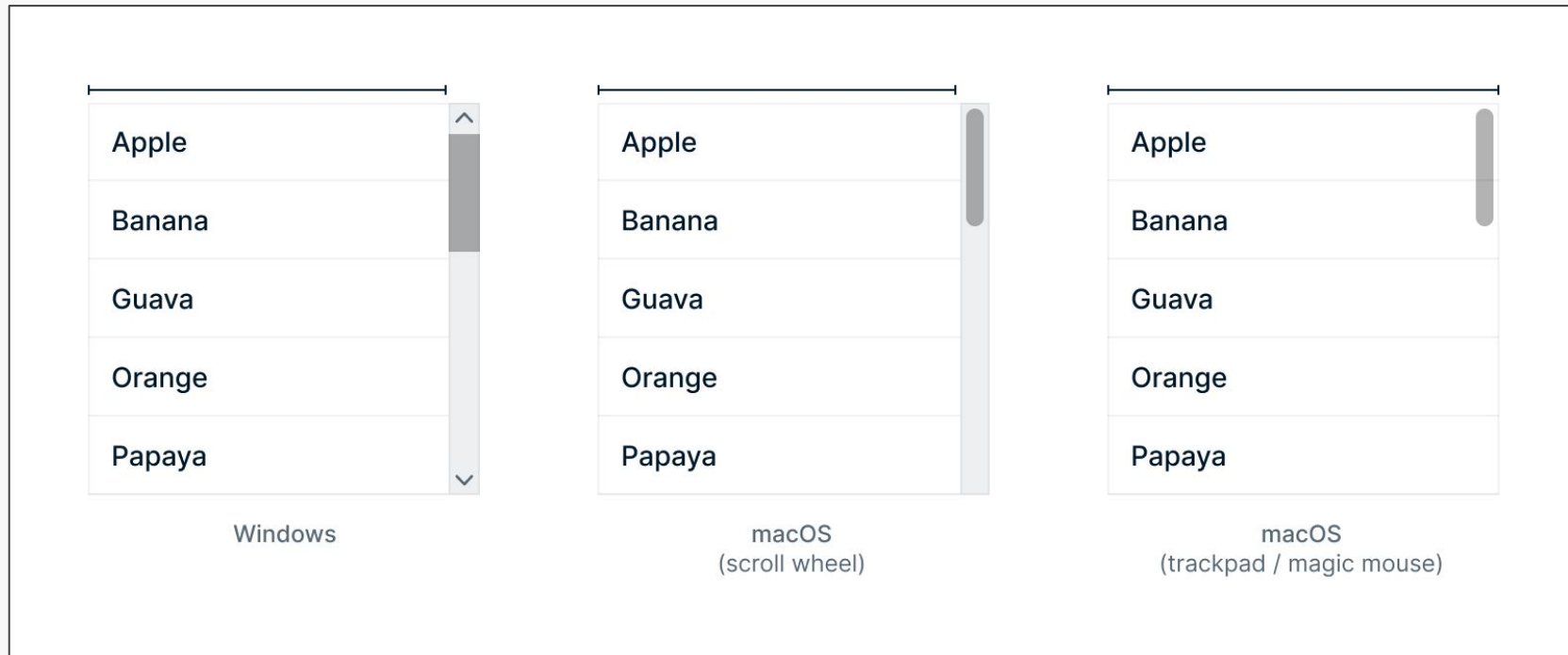


Figura: [Building like it's 1984: Scrollbars in web applications - Height](#)

CSS Tools: Reset CSS

The goal of a reset stylesheet is to reduce browser inconsistencies in things like default line heights, margins and font sizes of headings, and so on. The general reasoning behind this was [discussed in a May 2007 post](#), if you're interested. Reset styles quite often appear in CSS frameworks, and the original "meyerweb reset" found its way into [Blueprint](#), among others.

The reset styles given here are intentionally very generic. There isn't any default color or background set for the `body` element, for example. I don't particularly recommend that you just use this in its unaltered state in your own projects. It should be tweaked, edited,

[Post Archives](#)

[CSS Work](#)

[Toolbox](#)

[Writing](#)

[Speaking](#)

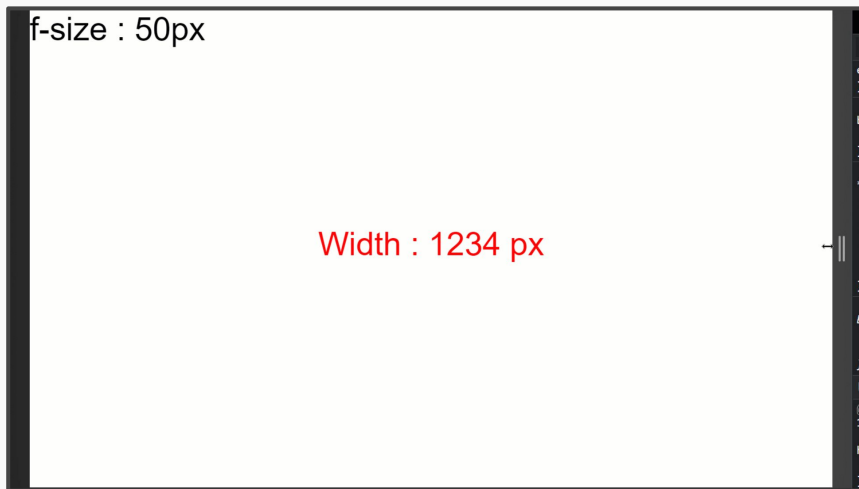
[Other Stuff](#)

[About this site](#)

Unidades de medida en CSS

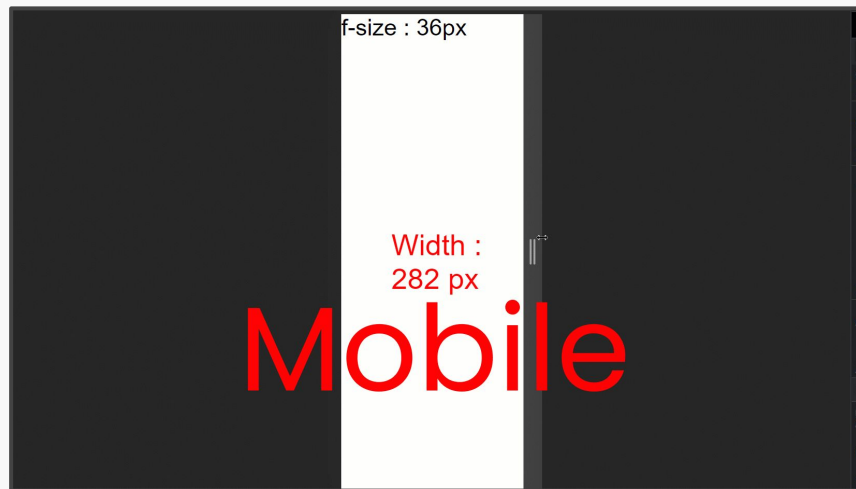
Unidades absolutas

PX



Unidades relativas

REM, EM, VW, VH



Unidades de medida en CSS

	body { font-size: 100%; }	body { font-size: 120%; }
font-size: 1em	The quick brown fox	The quick brown
font-size: 12pt	The quick brown fox	The quick brown fox
font-size: 16px	The quick brown fox	The quick brown fox
font-size: 100%	The quick brown fox	The quick brown

© KyleSchaetter.com

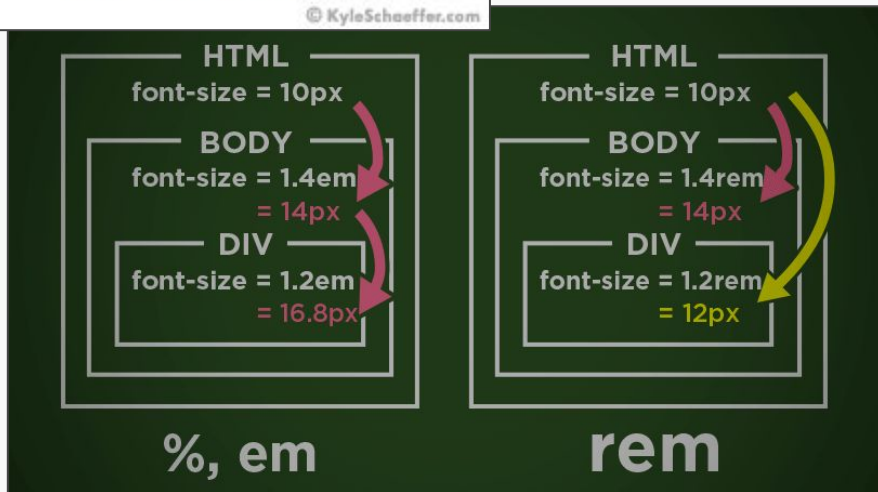


Figura: [A deep study on CSS Units](#)

Unidades de medida en CSS

SYMBOL	CLASS NAME	REMS	PX	EXAMPLE
	<code>.text-747</code>	2.5rem	40px	Good, cheap flights still exist. Here's how you can find them.
	<code>.text-777</code>	2rem	32px	Want to take your pet on your next international vacation?
	<code>.text-787</code>	1.5rem	24px	How much are you willing to pay for a few extra inches of legroom?
	<code>.text-767</code>	1.25rem	20px	Congress backs F.A.A measure but only with a few new traveler protections
	<code>.text-757</code>	1rem	16px	When being separated from your luggage is a good thing
	<code>.text-crj200</code>	1rem	16px	QUARANTINED AIRCRAFT AND SICK PASSENGERS ISN'T AS SCARY AS IT INITIALLY SOUNDS

Figura: [Why should REM be used instead of PX for UI coding?](#)

Aprende unidades CSS - Em, Rem, VH y VW con ejemplos de códigos:

Artículo describiendo y explicando las distintas unidades de medida en CSS

Search 800+ tutorials

freeCodeCamp (A)

Foro

Donar

15 DE SEPTIEMBRE DE 2021 / #CSS

Aprende unidades CSS - Em, Rem, VH y VW con ejemplos de códigos



Julio Vargas

CSS Units Explained

REM EM

VW VH



Aprende unidades CSS - Em, Rem, VH y VW con ejemplos de códigos:

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Julio Vargas

CSS Units Explained

REM EM

VW VH



Agenda

✨ CSS moderno

Frameworks y librerías

Design systems

CSS (presentación)

CSS (*Cascading Style Sheets*) es un **lenguaje especializado para diseño gráfico** (tampoco es lenguaje de programación) que define el “estilo”: indica cómo debe verse un elemento particular

Es un complemento a HTML, para implementar la presentación, visualización y diseño de páginas web. Para esto, ocupamos reglas CSS:

```
h1 {  
  color: green;  
}
```

Profesor: Antonio Ossa Guerra

Clases

1. Bienvenida al curso
2. De la URL al servidor
3. HTML y CSS
4. JavaScript
5. ...

Brevísima historia de CSS

- 1994 ● Creación de The World Wide Web Consortium ([W3C](#))
- 1996 ● **CSS1** Primera especificación de la W3C, incluyendo propiedades sobre *fonts*, colores, atributos de texto, y alineamiento y posicionamiento
- 1998 ● **CSS2** Superset de CSS1 agregando distintos tipos de posicionamiento, los conceptos de z-index y media types, y más
- 1999+ ● **CSS3** Se divide en “módulos” que agregan o extienden funcionalidad definida en CSS2. Estos siguen en trabajo y desarrollo al día de hoy
- 2007+ ● **CSS4** No hay una sola versión porque el *spec* está dividido en varios módulos con distintos niveles (ej. Flexbox)
- 2012 ● **CSS Flexbox** estable y soportado por navegadores
- 2016 ● **CSS Grid** estable y soportado por navegadores
- ... ● Nuevos módulos y *features* publicados constantemente

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Variables en CSS

Las variables en CSS (*custom properties*) son entidades que definen valores específicos que se pueden reutilizar en un documento. Existe sintaxis para declarar y para usar variables:

```
section {  
  --main-bg-color: brown;  
}
```

El selector determina el *scope*

```
:root {  
  --main-bg-color: brown;  
}
```

:root es la pseudo clase global

```
details {  
  background-color: var(--main-bg-color);  
}
```

Para usar el valor de una variable usamos var

CSS anidado

Si bien no agrega un comportamiento nuevo, CSS *nesting* permite escribir los estilos de una manera que es más fácil de leer, más modular y más mantenible, reduciendo la cantidad de selectores utilizados. Esta es una idea inspirada en el preprocesador [Sass](#)

```
/* Without nesting selector */
parent {
  /* parent styles */
  child {
    /* child of parent styles */
  }
}
```

```
/* With nesting selector */
parent {
  /* parent styles */
  & child {
    /* child of parent styles */
  }
}
```

```
/* the browser parses both of these as */
parent {
  /* parent styles */
}
parent child {
  /* child of parent styles */
}
```

Ambas escrituras son equivalentes a estos selectores

Selectores modernos

Existen una gran cantidad de combinadores/separadores y selectores:

Combinators and separators

- `+` ([Next-sibling combinator](#))
- `>` ([Child combinator](#))
- `||` ([Column combinator](#))
- `~` ([Subsequent sibling combinator](#))
- `" "` ([Descendant combinator](#))
- `|` ([Namespace separator](#))
- `,` ([Selector list](#))

Selectors

- `:active`
- `:any-link`
- `:autofill`
- `:blank`
- `:buffering`
- `:checked`
- `:current`
- `:default`
- `:defined`
- `:dir()`
- `:disabled`
- `:empty`
- `:enabled`

CanIUse: Sitio para evaluar disponibilidad de features en distintos navegadores

[Home](#)[News](#)

March 23, 2025 - New feature: View Transitions (cross-document)

[Compare browsers](#)[About](#)

Can I use

[?](#) [⚙ Settings](#)[Index of features](#)[▼ Filter features](#)

Latest features

- [View Transitions \(cross-document\)](#)
- [WebAssembly Import/Export of Mutable Globals](#)
- [WebAssembly Threads and Atomics](#)
- [WebAssembly Multi-Value](#)
- [WebAssembly BigInt to i64 conversion in JS API](#)

Most searched features

1. [WebP image format](#)
2. [Flexbox](#)
3. [CSS Grid](#)
4. [gap property for Flexbox](#)
5. [dvh \(Small, Large, and Dynamic viewport units\)](#)

Test a feature

Our partnership with [BrowserStack](#) now lets you test your website for compatibility across 2,000+ real browsers and devices.

Test on:

- [IE 11](#)
- [Safari on iPhone 14](#)
- [Safari 17](#)
- [Chrome on Galaxy S23](#)

Did you know?

If a feature you're looking for is not available on

Third party tools

- [The CanIUse Embed](#) - Add support tables to

Browser scores

[Current version](#) [Dev version](#)

Agenda

CSS moderno



Flexbox y Grid

Animaciones y similares

Frameworks y librerías

Design systems

Grid y Flexbox

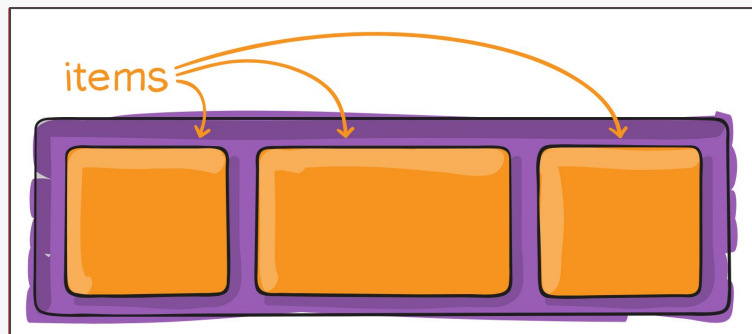
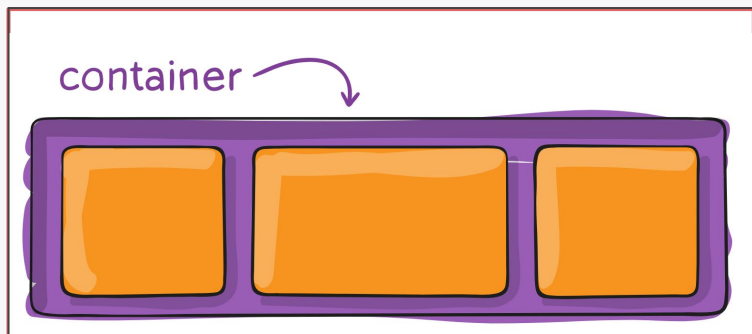
No existe una única manera o razón para elegir entre CSS Grid o CSS Flexbox más que el uso del criterio de quien programa. Cada uno fue diseñado con una fortaleza en mente:



Figura: [*Grid for layout, Flexbox for components*](#)

CSS Flexbox

El módulo “Flexbox Layout” apunta a proveer una forma más eficiente de distribuir y alinear elementos dentro de un contenedor incluso cuando su tamaño es desconocido o dinámico.



Guía completa para Flexbox | CSS Tricks



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LINKS

GUIDES

ALM

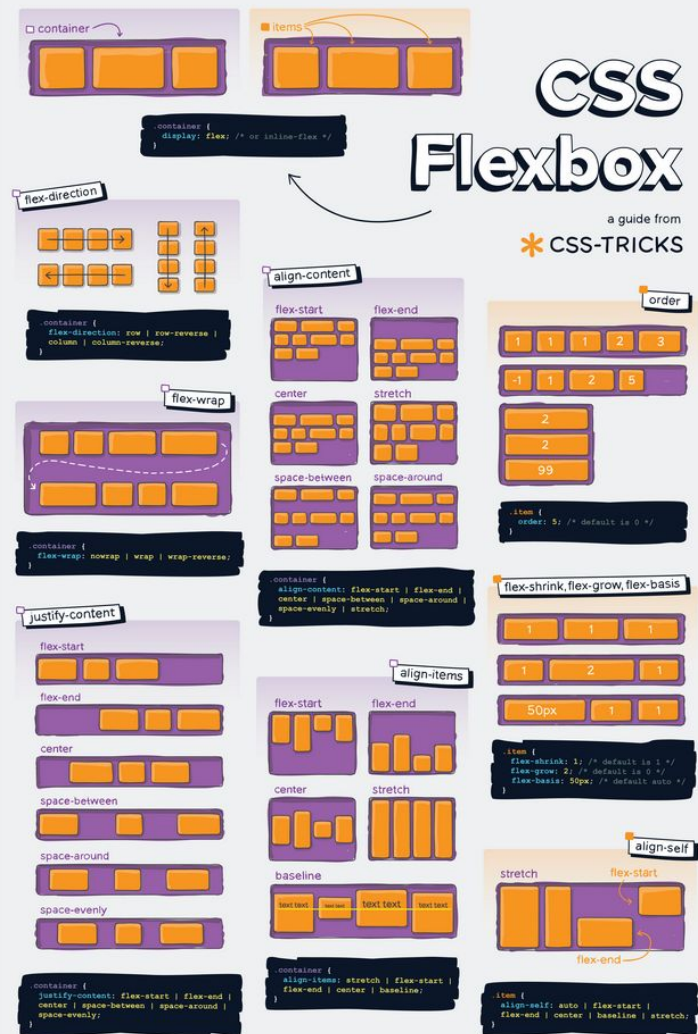
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CSS Flexbox Layout Guide



Chris Coyier on Aug 12, 2024

Our comprehensive guide to CSS flexbox layout. This complete guide explains everything about flexbox, focusing on all the different possible properties for the parent element (the flex container) and the child elements (the flex items). It also includes history, demos, patterns, and a browser support chart.



An Interactive Guide to Flexbox

Filed under **CSS** *on* November 22nd, 2022. *Last updated on* August 24th, 2024.

Flexbox is a remarkably powerful layout mode. When we *truly* understand how it works, we can build dynamic layouts that respond automatically, rearranging themselves as-needed.

For example, check this out:

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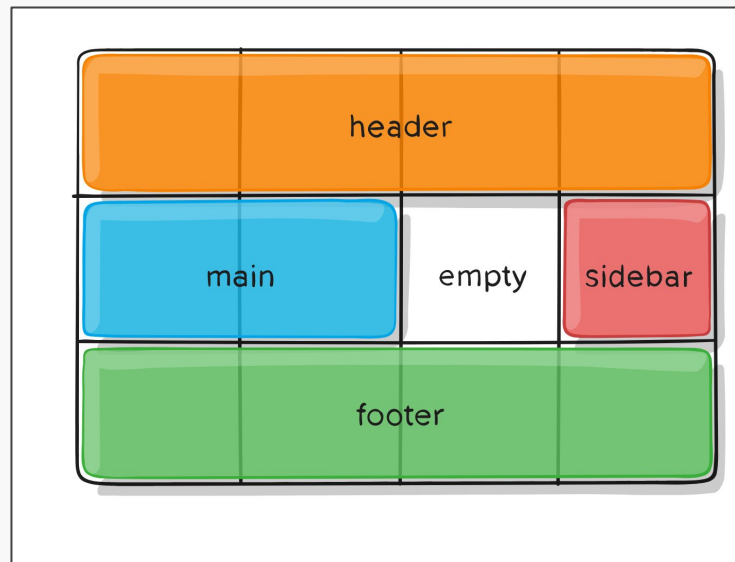
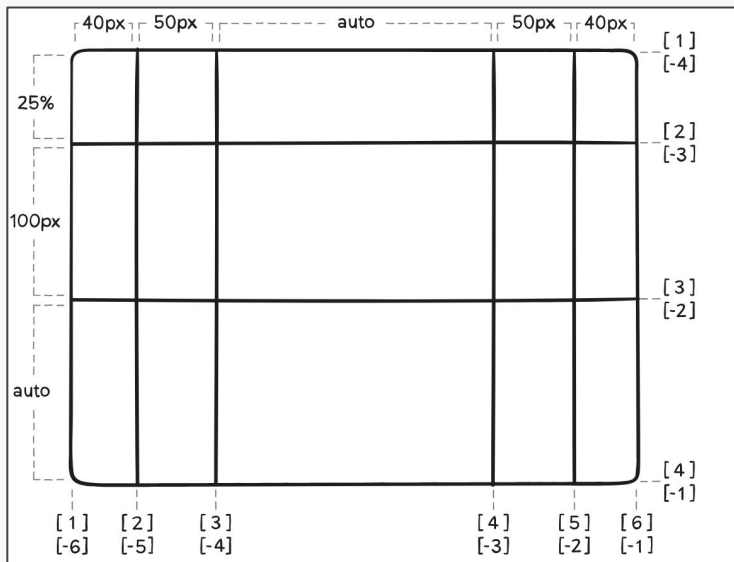
[Introduction to Flexbox](#)

[Flex direction](#)

[Alignment](#)

CSS Grid

El módulo “Grid Layout” dispone de un *layout* bidimensional, otorgando control sobre las celdas de la grilla con elementos que abarquen desde una sola celda hasta áreas (espacio entre 4 líneas de grillas).



CSS Grid Layout Guide: Guía completa para Grid | CSS Tricks

* CSS-TRICKS

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CSS Grid Layout Guide



Chris House on Aug 12, 2024

Our comprehensive guide to CSS grid, focusing on all the settings both for the grid parent container and the grid child elements.

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Part 1: [Introduction](#)

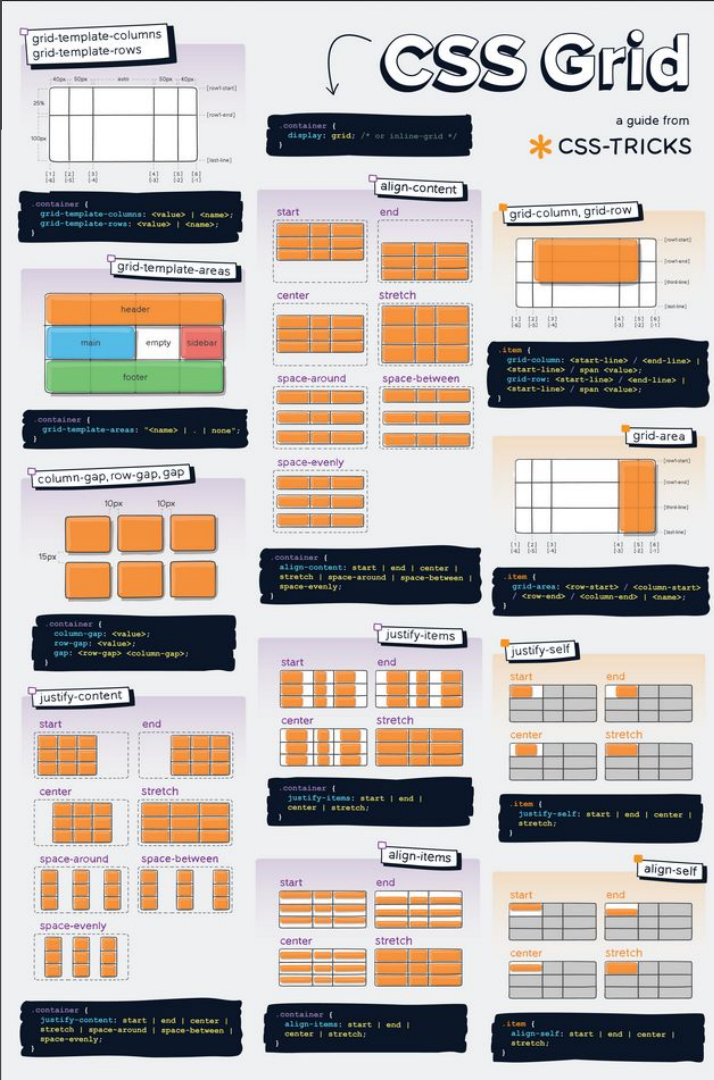
Part 2: [Basics](#)

Get the po

Reference thi

lot? Here's a h

image you can



An Interactive Guide to CSS Grid • Josh W. Comeau:

Artículo para experimentar con Grid y sus propiedades

[Categories](#)[Courses](#)[Goodies](#)[About](#)

An Interactive Guide to CSS Grid

Filed under **CSS** *on* November 21st, 2023. *Last updated on* September 13th, 2024.

CSS Grid is one of the most amazing parts of the CSS language. It gives us a ton of new tools we can use to create sophisticated and fluid layouts.

It's also *surprisingly complex*. It took me quite a while to truly become comfortable with CSS Grid!

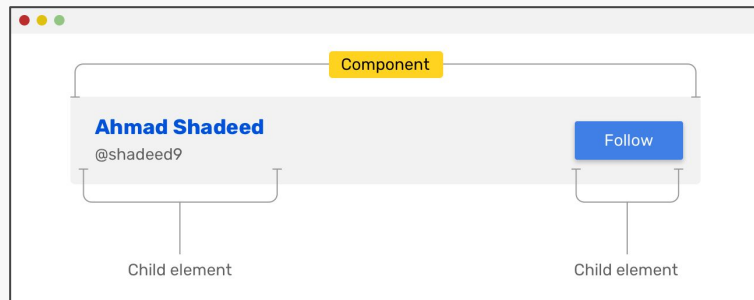
TABLE OF CONTENTS

[Introduction](#)[Mental model](#)[Grid flow](#)

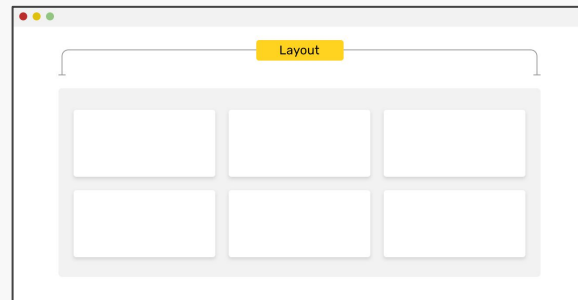
¿Cómo decidir cuál usar?

Si los componentes se muestran en línea (1 dimensión) usamos Flexbox, pero si tiene filas y columnas utilizaremos Grid

Además, hay que considerar la responsividad (cómo se “mueven” los elementos en diferentes *viewports*)



Elementos alineados ➡ **CSS Flexbox**



Filas y columnas ➡ **CSS Grid**

Agenda

CSS moderno

Flexbox y Grid



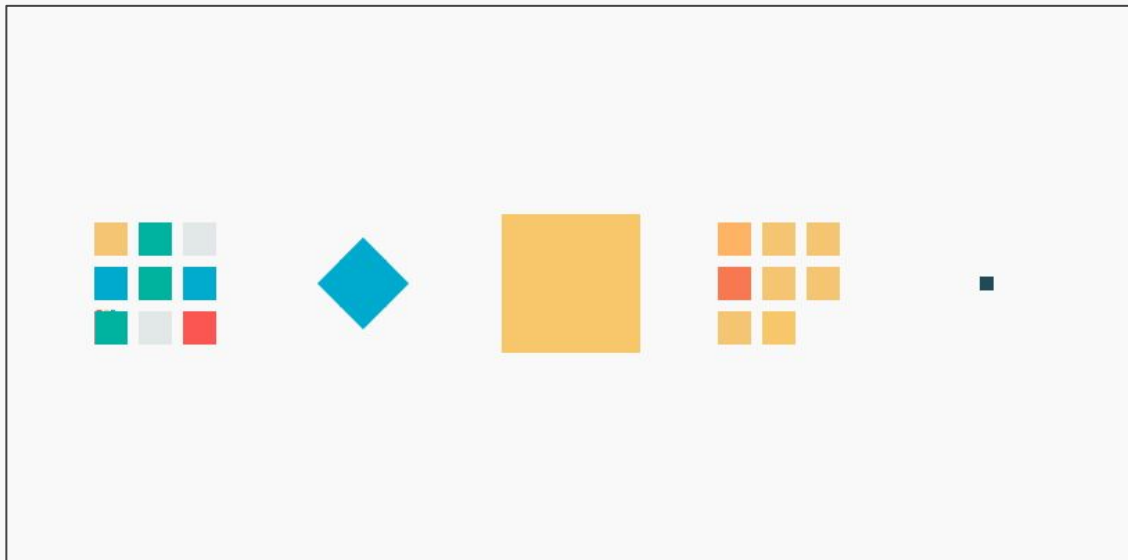
Animaciones y similares

Frameworks y librerías

Design systems

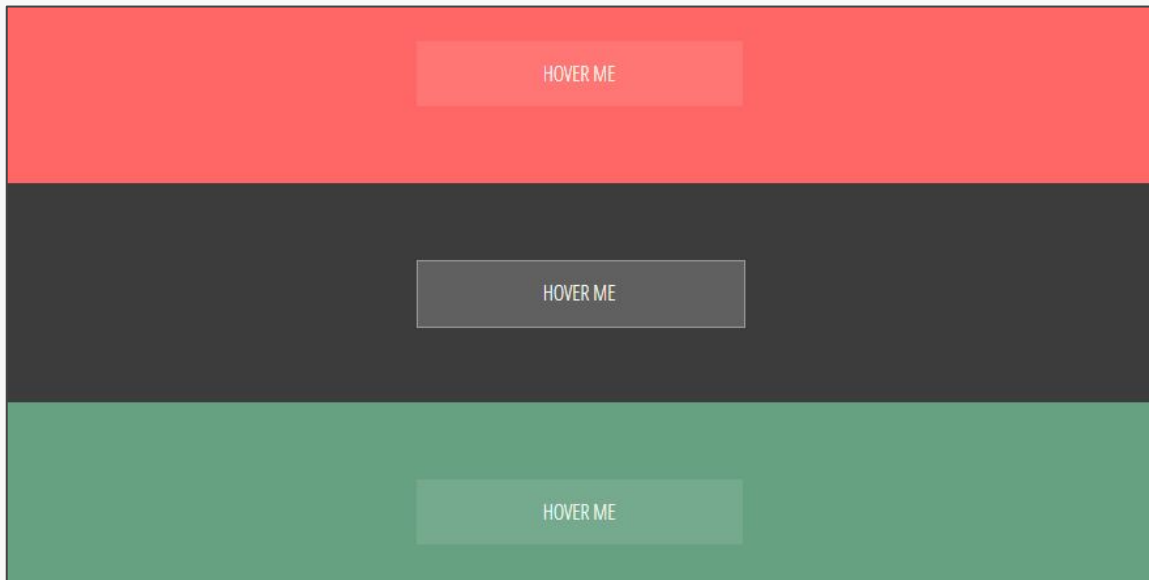
Animaciones en CSS

Las animaciones en CSS permiten animar la transición desde un estilo CSS (definido por ciertas reglas) a otro, entregando dinamismo en distintos elementos, donde tendremos control tanto a nivel de la duración de la animación como en los cambios de estilo



Propiedad `transition`

Usamos la propiedad `transition` para indicar cómo ocurre la transición entre dos estados de un elemento, por ejemplo, entre el estado “por defecto” y al hacer *hover* sobre este:



Propiedad transition

Try it

CSS Demo: transition

RESET

transition: margin-right 2s;

transition: margin-right 2s .5s;

transition: margin-right 2s ease-in-out;

transition: margin-right 2s ease-in-out

< >

transition: margin-right 2s, color 1s;

transition: all 1s ease-out;

Hover to see
the transition.

CSS

We include two `<time>` values. In the `transition` shorthand, the first `<time>` value is the `transition-duration`. The second time value is the `transition-delay`. Both default to `0s` if omitted.

CSS

  Play

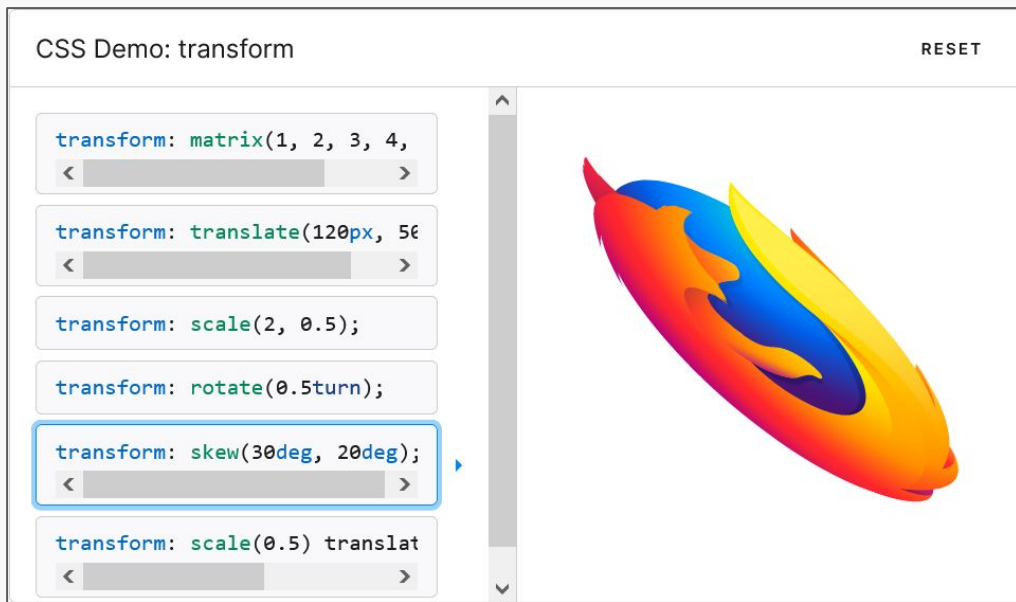
```
.target {  
  font-size: 2rem;  
  background-color: palegoldenrod;  
  transition: background-color 2s 500ms;  
}  
  
.target:hover {  
  background-color: darkorange;  
}
```

 Play

Hover over me

Propiedad transform

Usamos la propiedad transform para rotar, escalar, girar, mover o trasladar un elemento, modificando el espacio de coordenadas del modelo en CSS:



Propiedad animation

Usamos la propiedad `animation` para definir los detalles de los tiempos (*timing*) de la animación descrita. Podemos controlar múltiples aspectos de la animación:

```
1 /* Versión verbosa */
2 p {
3     animation-duration: 3s;
4     animation-name: slidein;
5     animation-iteration-count: infinite;
6     animation-direction: alternate;
7 }
8
9 /* Solo con animation */
10 p {
11     animation: 3s infinite alternate slidein;
12 }
13
```

Uso de @keyframes

Con @keyframes definimos los estilos que se aplicarán en nuestra animación y en qué momento de esta ocurrirán utilizando porcentajes

```
1 ~ .element {  
2   animation: pulse 5s infinite;  
3 }  
4  
5 ~ @keyframes pulse {  
6   0% {  
7     background-color: #001F3F;  
8   }  
9   100% {  
10    background-color: #FF4136;  
11  }  
12 }  
13
```

Using CSS animations - CSS: Cascading Style Sheets | MDN:

Guía en MDN para introducirse en el uso de animaciones con CSS

Filter

Types

Guides

Animations

Using CSS animations

Backgrounds and Borders

Box alignment

Box model

Colors

Columns

Conditional rules

Using CSS animations

CSS animations make it possible to animate transitions from one CSS style configuration to another. Animations consist of two components: a style describing the CSS animation and a set of keyframes that indicate the start and end states of the animation's style, as well as possible intermediate waypoints.

There are three key advantages to CSS animations over traditional script-driven animation techniques:

1. They're easy to use for basic animations; you can create them without even having to know JavaScript.
2. The animations run well, even under moderate system load. Simple animations can often perform poorly in JavaScript. The rendering engine can use frame-skipping and other techniques to keep the performance as

In this article

Configuring an animation

Defining an animation sequence using keyframes

Using the animation shorthand

Setting multiple animation property values

Examples

See also

SVG guides - Scalable Vector Graphics:

Guías en MDN para trabajar con SVGs dentro de documentos

Filter

SVG

Tutorials

► Introducing SVG from scratch

Guides

SVG effects for HTML

Content type

Linking

Namespaces crash course

Scripting

SVG animation with SMIL

SVG guides

This page lists guides for SVG.

[Applying SVG effects to HTML content](#)

Modern browsers support using [SVG](#) within [CSS](#) styles to apply graphical effects to HTML content.

[Content type](#)

SVG makes use of a number of data types. This article lists these types along with their syntax and descriptions of what they're used for.

[Linking](#)

The `target` attribute on the SVG `a` element doesn't work in Mozilla Firefox

In this article

See also

Videos y recursos recomendados

- [How to Properly Layout A Website \(For Beginners\)](#) by [The Website Architect](#)
- [The Easy Way to Design Top Tier Websites](#) by [Sajid](#)
- [world's shortest Figma course](#) by [Juxtopposed](#)
- [Make Awesome SVG Animations with CSS // 7 Useful Techniques](#) by [Fireship](#)
- [Subtle, yet Beautiful Scroll Animations](#) by [Beyond Fireship](#)
- <https://youtu.be/ymuBowcODVU> by [Lun Dev](#)
- [Animating SVG with CSS](#) by CSS Tricks

Agenda

CSS moderno



Frameworks y librerías

Design systems

Bootstrap: Framework para desarrollo de front-end con estilos listos



[Docs](#) [Examples](#) [Icons](#) [Themes](#) [Blog](#)

🔍 Search

CTRL K



v5.3 ▾



Get Security Updates for Bootstrap 3 & 4 →



Build fast, responsive sites with Bootstrap

Powerful, extensible, and feature-packed frontend toolkit. Build and customize with Sass, utilize prebuilt grid system and components, and bring projects to life with powerful JavaScript plugins.

\$ npm i bootstrap@5.3.5 📄

📖 [Read the docs](#)

Currently **v5.3.5** · [Download](#) · [All releases](#)

Bootstrap: Framework para desarrollo de front-end con estilos listos



Docs Examples Icons Themes Blog

Q Search

CTRL K



v5.3



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Bootstrap Example</title>
    <!-- Bootstrap CSS via CDN -->
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.5/dist/css/bootstrap.min.css"
      rel="stylesheet"
      integrity="sha384-Sg0Ja3DmI69IuzQ2PVdRZhWq+dy64/BUTbMJw1MZ8t5HZApCtHrRKUc4W0kG879m7"
      crossorigin="anonymous"
    />
  </head>
  <body>
    <div class="container mt-5">
      <h1 class="text-primary">Hello, Bootstrap!</h1>
      <button class="btn btn-success">Click Me</button>
    </div>
    <!-- Bootstrap JS via CDN -->
    <script
      src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.5/dist/js/bootstrap.min.js"
      integrity="sha384-k6d4wzSIapyDyv1kpU366/PK5hCdSbCRGRCMv+ep1C"
      crossorigin="anonymous"
    ></script>
  </body>
</html>
```

Hello, Bootstrap!

Click Me



Include via CDN

When you only need to include Bootstrap's compiled CSS or JS, you can use [jsDelivr](#). See it in action with our simple [quick start](#), or [browse the examples](#) to jumpstart your next project. You can also choose to include Popper and our JS [separately](#).

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.5/dist/css/b
```

```
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.5/dist/js/b
```



[Playground](#)

[Install](#)

[Learn Sass](#)

[Blog](#)

[Documentation](#)

[Get Involved](#)

[CTRL](#) [K](#)

CSS with superpowers

Sass is the most mature, stable, and powerful professional grade CSS extension language in the world.



Sass: Extensión del lenguaje CSS con elementos adicionales

```
@use 'sass:color';
```

```
$primary-color: #3498db;  
$padding: 1rem;
```

```
.card {  
  background: color.adjust($primary-color, $lightness: 20%);  
  border: 1px solid color.adjust($primary-color, $lightness: -10%);  
  padding: $padding;  
  
  .card-title {  
    font-size: 1.5rem;  
    color: $primary-color;  
  }  
}
```

When you install Sass on the command line, you'll be able to run the sass executable to compile .sass and .scss files to .css files. For example:

```
sass source/stylesheets/index.scss build/stylesheets/index.css
```

Sass is the most mature, stable, and powerful professional grade CSS extension language in the world

Current Releases: [Dart Sass 1.86.0](#)

```
.card {  
  background: rgb(138.6359832636, 195.9581589958, 234.3640167364);  
  border: 1px solid rgb(33.1380753138, 125.1882845188, 186.8619246862);  
  padding: 1rem;  
}  
.card .card-title {  
  font-size: 1.5rem;  
  color: #3498db;  
}
```

```
/*# sourceMappingURL=styles_con_scss.css.map */
```

Sass: Extensión del lenguaje CSS con elementos adicionales

Free Palestine

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Sass Card Example</title>
    <!-- Link to the compiled CSS from styles_con_scss.scss -->
    <link rel="stylesheet" href="styles_con_scss.css" />
  </head>
  <body>
    <div class="card">
      <h2 class="card-title">Hello, Sass Card!</h2>
      <p>This card is styled using the generated CSS file.</p>
    </div>
  </body>
</html>
```

Sass is the most mature, stable, and powerful professional grade CSS extension language in the world.

Learn Sass

Blog

Documentation

Get Involved

Search

erpowers

Hello, Sass Card!

This card is styled using the generated CSS file.

Current Releases: [Dart Sass 1.86.0](#) [LibSass](#)

Less CSS: Lenguaje de *scripting* que apunta a extender CSS

[Overview](#)[Using Less.js](#)[Functions](#)[In-Depth Guide](#)[Tools](#)[About](#)[GitHub](#) ▾

It's CSS, with just a little more.

Use with Node.js:

```
npm install -g less  
> lessc styles.less styles.css
```

Or the browser:

```
<link rel="stylesheet/less" type="text/css" href="styles.less" />  
<script src="https://cdn.jsdelivr.net/npm/less" ></script>
```

Or try [the online playground](#) ! 

Less CSS: Lenguaje de *scripting* que apunta a extender CSS

```
@primary-color: #e74c3c;
@radius: 4px;
@padding: 8px;

.button {
  background-color: @primary-color;
  border-radius: @radius;
  padding: @padding 2 * @padding;

  &:hover {
    background-color: lighten(@primary-color, 10%);
  }
}
```

Use with Node.js:

```
npm install -g less
```

```
> lessc styles.less styles.css
```

```
> lessc
```

```
.button {
  background-color: #e74c3c;
  border-radius: 4px;
  padding: 8px 16px;
}

.button:hover {
  background-color: #ed7669;
}
```

```
<link rel="stylesheet/less" href="styles.css">
```

```
<script src="https://cdnjs.cloudflare.com/ajax/libs/less.js/2.7.3/less.min.js">
```

Or try [the online playground](#) ! 

Less CSS: Lenguaje de *scripting* que apunta a extender CSS

[Overview](#)[Using Less.js](#)[Functions](#)[In-Depth Guide](#)[Tools](#)[About](#) [GitHub](#) ▾

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Button Preview</title>
    <!-- Link to your CSS file -->
    <link rel="stylesheet" href="styles_con_less.css" />
  </head>
  <body>
    <!-- Example usage of the .button class -->
    <button class="button">Click Me</button>
  </body>
</html>
```

```
> lessc styles.less styles.css
```

Or the browser:

```
<link rel="stylesheet/less" type="text/css" href="styles.less" />
<script src="https://cdn.jsdelivr.net/npm/less" ></script>
```

Or try [the online playground](#) ! 

Click Me

Tailwind CSS: Framework de CSS con clases pre-definidas



v4.0 ▾

Q Ctrl K

Docs

Blog

Showcase

Plus



text-8xl text-white tracking-tighter text-balance

Rapidly build modern websites without ever leaving your HTML.

text-lg text-white font-medium

A utility-first CSS framework packed with classes like `flex`, `pt-4`, `text-center` and `rotate-90` that can be composed to build any design, directly in your markup.

Get started

Q Quick search

Ctrl K

Tailwind CSS: Framework de CSS con clases pre-definidas



v4.0

Q Ctrl K

Docs

Blog

Showcase

Plus



```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="UTF-8" />
    <title>Tailwind Example</title>
    <!-- Tailwind via CDN for quick prototyping -->
    <script src="https://cdn.jsdelivr.net/npm/@tailwindcss/browser@4"></script>
  </head>
  <body class="bg-gray-100 flex items-center justify-center h-screen">
    <div
      class="p-6 max-w-sm mx-auto bg-white rounded-xl shadow-md flex items-center space-x-4"
    >
      <div>
        <h2 class="text-xl font-bold text-indigo-600">Hello, Tailwind!</h2>
        <p class="text-gray-500">Quick utility-first styling.</p>
      </div>
    </div>
  </body>
</html>
```

Hello, Tailwind!

Quick utility-first styling.

Get started with Tailwind CSS

Tailwind CSS works by scanning all of your HTML files, JavaScript components, and any other templates for class names, generating the corresponding styles and then writing them to a static CSS file.

It's fast, flexible, and reliable — with zero-runtime.

Installation

[Using Vite](#)

[Using PostCSS](#)

[Tailwind CLI](#)

[Framework Guides](#)

[Play CDN](#)

Get started

Q Quick search

Three.js: Librería para gráficos 3D en JavaScript

three.js r176

Learn

examples

documentation

devtools

gpt

Play

editor

nodes

Community

questions

discord

forum

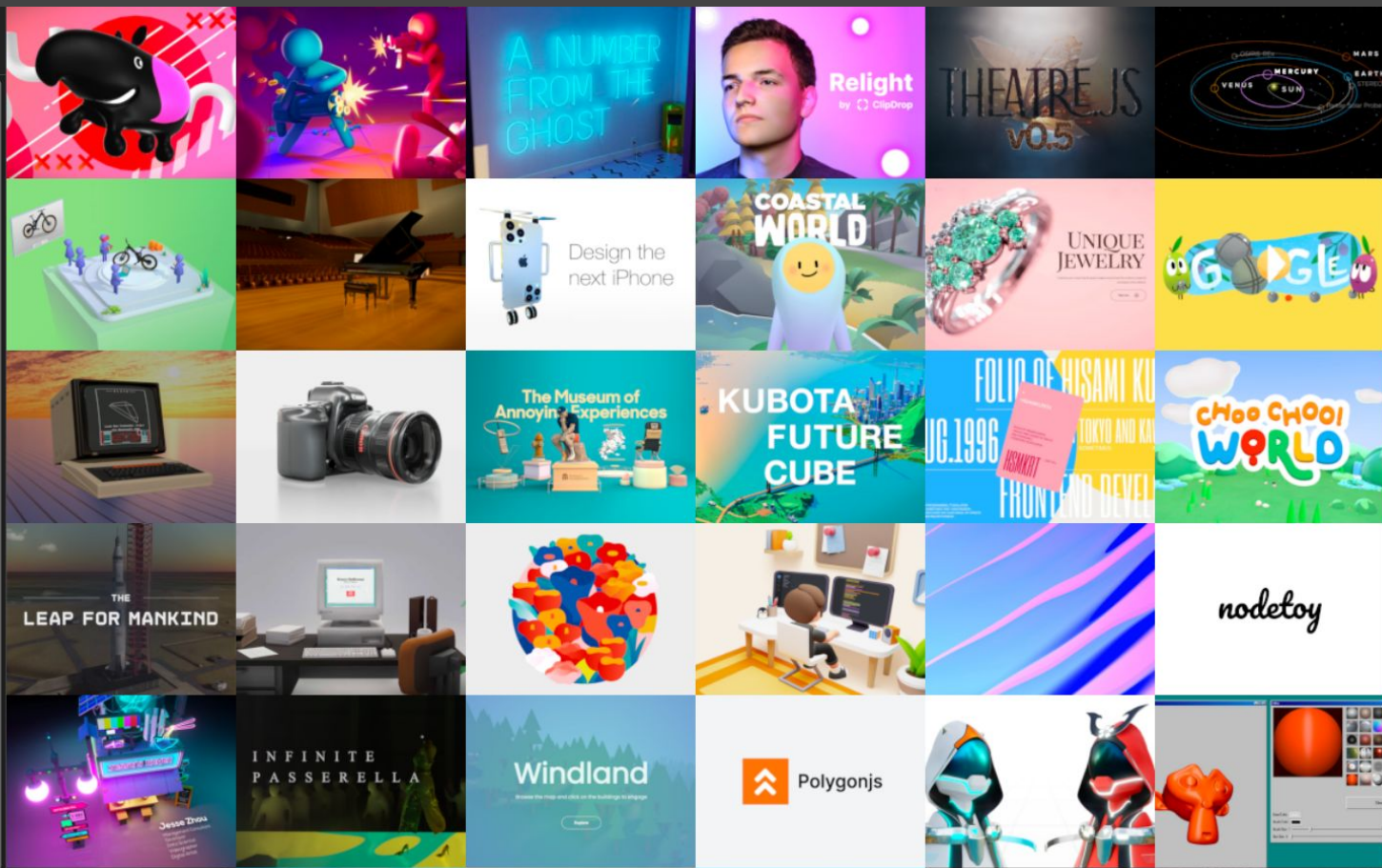
twitter

Code

github

download

Resources



Three.js: Librería para gráficos 3D en JavaScript

```
import * as THREE from 'three';

// Set up the scene, camera, and renderer
const scene = new THREE.Scene();
const camera = new THREE.PerspectiveCamera(
  75,
  window.innerWidth / window.innerHeight,
  0.1,
  1000
);

const renderer = new THREE.WebGLRenderer();
renderer.setSize(window.innerWidth, window.innerHeight);
document.body.appendChild(renderer.domElement);

// Create a cube with different colors for each face
const geometry = new THREE.BoxGeometry();
const materials = [
  new THREE.MeshBasicMaterial({ color: 0xff0000 }), // Red
  new THREE.MeshBasicMaterial({ color: 0x00ff00 }), // Green
  new THREE.MeshBasicMaterial({ color: 0x0000ff }), // Blue
  new THREE.MeshBasicMaterial({ color: 0xffff00 }), // Yellow
  new THREE.MeshBasicMaterial({ color: 0xff00ff }), // Magenta
  new THREE.MeshBasicMaterial({ color: 0x00ffff }), // Cyan
];
const cube = new THREE.Mesh(geometry, materials);
scene.add(cube);

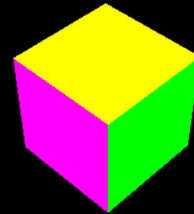
// Position the camera
camera.position.z = 5;

// Animation loop
function animate() {
  requestAnimationFrame(animate);

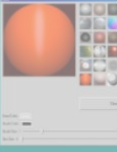
  // Rotate the cube
  cube.rotation.x += 0.01;
  cube.rotation.y += 0.01;

  renderer.render(scene, camera);
}
animate();
```

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Minimal Three.js Example</title>
    <style>
      body { margin: 0; }
      canvas { display: block; }
    </style>
    <script type="importmap">
      {
        "imports": {
          "three": "https://cdn.jsdelivr.net/npm/three@0.176.0/build/three.module.js",
          "three/addons/": "https://cdn.jsdelivr.net/npm/three@0.176.0/examples/jsm/"
        }
      }
    </script>
  </head>
  <body>
    <script type="module" src="/main.js"></script>
  </body>
</html>
```



nodetoy



Agenda

CSS moderno

Frameworks y librerías

✨ *Design systems*

Design Systems

Un *design system* es un conjunto de estándares para manejar y construir un diseño a escala utilizando un “lenguaje común” y consistencia visual a través de múltiples páginas y canales. Algunos ejemplos:

- [Fintual Design System / Design Systems International](#)

Fuente y figura: [Design Systems 101](#)

sequence of events which includes: arriving at a facility, going to a specific building, then seeking a floor and room number. Relating the specific sign to a larger context will yield the best results. Also, categorize signs by functional types as a method of simplifying the overall signing task.

A sign should be thought of as a large-scale headline; therefore, language should be clear and concise. Brevity is desirable in order to communi-

cate quickly, especially to drivers of vehicles.

Placement of the sign is very important. The sign should be placed for optimum viewing distance. It is good to test these conditions by creating mockup signs out of photostats and inexpensive materials, and thus determine their effectiveness before fabricating the finished product.

Consider environmental factors when developing signage. Weather conditions should determine the material selected and the fabrication tech-

nique. Color should be chosen based on the type of Sun conditions which prevail; i.e., a dark background with reversed (white) letters will be more legible against a bright desert sky.

Use consistent message formats to create a uniform look and coordinated sign program. NASA signs should employ the flush left, ragged right format as demonstrated on these pages.



Design Systems International:

Es una consultora de diseño digital (instalada en Chile)

DESIGN SYSTEMS
INTERNATIONAL

WORK IDEAS ABOUT TEAM

Build distinctive experiences
at the speed modern
business requires

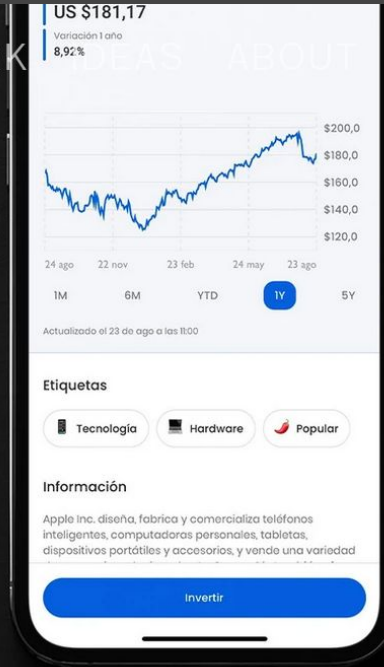


Fintual Design System / Design Systems International:

Case Study: Fintual - Connecting product and brand with a new design system

DESIGN SYSTEMS
INTERNATIONAL

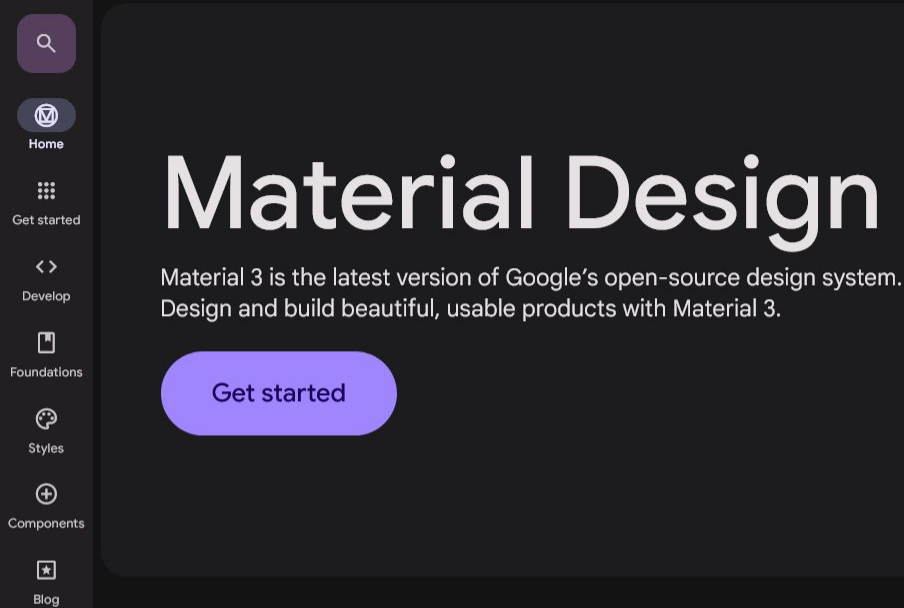
WORK IDEAS ABOUT TEAM



Fintual
Connecting product and brand
with a new design system

Material Design:

Es un *design system* que incluye guías y componentes para diseñar interfaces

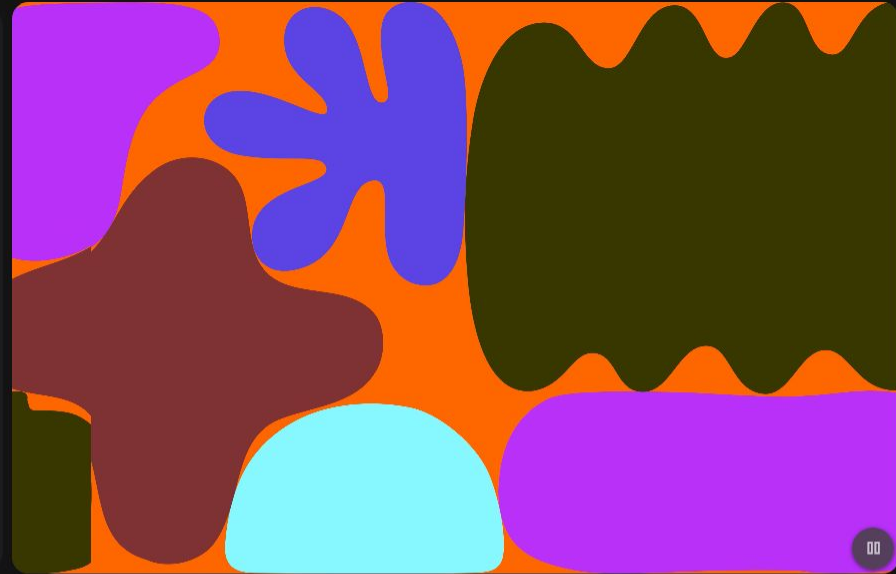


The header of the Material Design website features a dark grey background. On the left is a vertical sidebar with icons and labels: a magnifying glass for search, a home icon labeled 'Home', a grid icon labeled 'Get started', a double arrow icon labeled 'Develop', a book icon labeled 'Foundations', a paint palette icon labeled 'Styles', a plus icon labeled 'Components', and a star icon labeled 'Blog'. The main content area on the right has the title 'Material Design' in large white font. Below it, a paragraph states: 'Material 3 is the latest version of Google's open-source design system. Design and build beautiful, usable products with Material 3.' A purple rounded button with the text 'Get started' is positioned below the paragraph.

Material Design

Material 3 is the latest version of Google's open-source design system. Design and build beautiful, usable products with Material 3.

Get started



News & launches



Brand Colors

The brand colors are reserved for only the Redis logo.



Core Colors

Core colors are used in the majority of our web pages, presentations, and other collateral.



Secondary Colors

Secondary colors are limited to small pops of colors, such as for our accent icons or in some of our illustrations.



Neutral Colors



Alert Colors



Typography

Aa Bb Cc

Graphik is our primary brand font and should be used for all headlines, sub-headlines, body copy, titles, calls-to-action, and labels.

However, when it comes to our Google Docs, or Google Slides/presentations, we use the universal font, Calibri.

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz0123456789
!@#\$%^&*()-+[]{};':/.,

Graphik Semibold

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz0123456789

Graphik Medium

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz0123456789

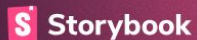
Graphik Regular

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz0123456789

Graphik Light

Redis Brand Guidelines:
Redis Brand Guidelines
(edición anterior, no actual)

Storybook: Herramienta para crear y organizar elementos de un *design system*

[Docs](#)[Addons](#)[Showcase](#)[Blog](#)[Visual Test](#)[Enterprise](#)[86,456](#)[Search docs](#)[CTRL K](#)

Build UIs without the grunt work

Storybook is a frontend workshop for building UI components and pages in isolation. Thousands of teams use it for UI development, testing, and documentation. It's open source and free.

[Get Started](#)[npm create storybook@latest](#)

33.61m

Installs per month

2282

Contributors

Version **8**

[Development](#)[Component testing](#)[Visual testing](#)[Documentation](#)

localhost:6006

CSSBattle: Sitio para intentar replicar imagenes con CSS y practicar



158 Online



Sign In / Sign Up



GET
STARTED



PLAY



Welcome to CSSBattle

The funnest multiplayer game with 300K+ web designers & developers. Replicate the target images using CSS - the shorter your code, the higher your score! Happy coding!

Sign In / Sign Up



Daily targets

A new target everyday for you to unwind. No leaderboards, no competition

View all daily targets

MAR 24

MAR 25 (TODAY)

TOMORROW

Agenda

CSS moderno

Frameworks y librerías

Design systems



¿Qué se viene pronto?

Próximas fechas importantes: clases y ayudantías

	Clases y ayudantías						
	Lunes	Martes	Miércoles	Jueves	Viernes	Sábado	Domingo
Semana 7 (14/4 - 20/4)		Fundamentos de front-end		Jueves Santo	Viernes Santo	Sábado Santo	Domingo de Resurrección
Semana 8 (21/4 - 27/4)		Charla Front-end (Florencia Valladares)		HOY			
Semana 9 (28/4 - 04/5)	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	
Semana 10 (05/5 - 11/5)		Lanzamiento proyecto (E0)		Post-receso y repaso			
Semana 11 (12/5 - 18/5)		Cookies y sessions		Autenticación y autorización			

Próximas fechas importantes: evaluaciones

	Evaluaciones						
	Lunes	Martes	Miércoles	Jueves	Viernes	Sábado	Domingo
Semana 7 (14/4 - 20/4)				Jueves Santo	Viernes Santo	Sábado Santo	Domingo de Resurrección
Semana 8 (21/4 - 27/4)	UC: Sin eval.	UC: Sin eval.	Fin T2 (API)	HOY			
Semana 9 (28/4 - 04/5)	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Semana de Receso	UC: Sin eval.
Semana 10 (05/5 - 11/5)	UC: Sin eval.	Inicio E0					
Semana 11 (12/5 - 18/5)					Fin E0	Inicio E1	

Clase 13

Diseño web en la práctica

IIC2513 - Tecnologías y Aplicaciones Web

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